VU AI HUB · LECTURE NOTES

Introduction to Psychology and its Methods

Year 1, September to October 2025

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2nd of September

Topic 1 - Genes and Evolution

Psychology is the scientific study of mind, brain and behavior ~ Gazzaniga, 2018

The genetic basis

In the 70's scientists claimed that the environment caused your behavior, but since the 90's they claimed that it was coming from your genes and nowadays they believe it's a combination of both

Nature (genes, biology, heredity etc.) vs. Nurture (environment etc.)

Chromosomes containing DNA are situated in the nucleus of a cell

- Within chromosomes you have DNA and within the DNA you have your genes

Genes are functional biochemical units of heredity that make up the chromosomes, they're the meaningful sections of the DNA molecule. They govern the cell's functioning by providing instructions for making proteins.

Genes typically are not solo players

Gene expression (whether a gene is turned 'on' or 'off') depends on methylation:

- a chemical reaction in the body in which a small molecule called a methyl group gets added to DNA, proteins, or other molecules
- it's an explanation of environmental influence on genes

Gene expression is controlled by the biochemical environment inside the cell, and that is influenced by the environment outside the cell, the timing in development, the overall environment, experience and behavior.

Genome is the set of complete instructions for making an organism, it contains all the genes in that organism. The human genome makes us human.

Genotype is an organism's specific set of genes

Phenotype is the overt traits and behaviors of an organism (determined by the genotype x environment)

Each gene is paired with another gene

Allele is one specific variant of a gene

- dominant
- recessive

Homozygote: alleles on locus are the same

Heterozygote: alleles on locus are different

Evolution by natural selection

The key mechanism of evolution is natural selection (Charles Darwin)

3 conditions:

- There is variation among individuals of populations
- Individuals with a certain trait survive and reproduce at higher rates than others
- The trait associated with this advantage is passed from parents to offspring

What matters is the survival of genes, not the survival of individuals (they're just the carriers)

The evidence for modern evolutionary theory comes from many sources

- The fossil record
- The resemblance between genomes of various organisms
- Pseudogenes (junk DNA - lacks regulatory elements)
- Distributions of species across the world
  - continental islands vs. oceanic islands

4th of September

Topic 2 - The Brain and the Nervous System

Studying the brain

19th century - Phrenology

- Bumps on the skull were interpreted in terms of personality traits

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Methods for studying the brain

- Clinical observation of patients with brain damage
- Experimental techniques
  - Invasise: animal studies (lesions, single-cell recordings etc.)
  - TMS (Transcranial Magnetic Simulation)
    - magnets interfere with electric fields, so by putting a magnetic coil on you head it interferes with your brain activity
- Other techniques
  - Electrophysiology
    - EEG (ERP - Event Related Potential Technique)
  - Brain Imaging

- (f)MRI scan (tells you where and when activity occurs)
- PET scan (tells you where activity occurs)

Building blocks of the nervous system

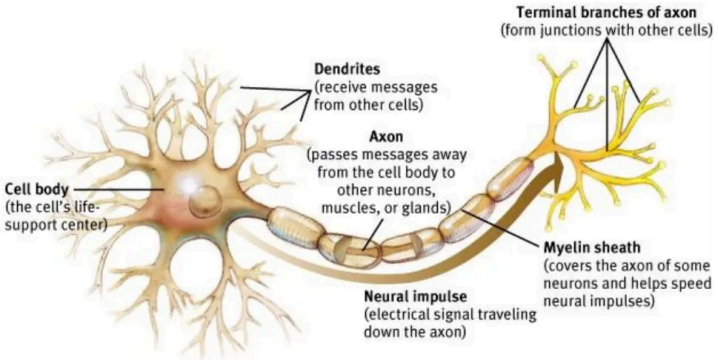
The nervous system is made up of two basic kinds of cells

- Glia
  - support functions like increasing and decreasing neurotransmitters
- Neurons
  - they typically allow you to talk, walk, eat etc.

Different types of neurons

- Sensory receptors
- Sensory (afferent) neurons
- Motor (efferent) neurons
- Interneurons

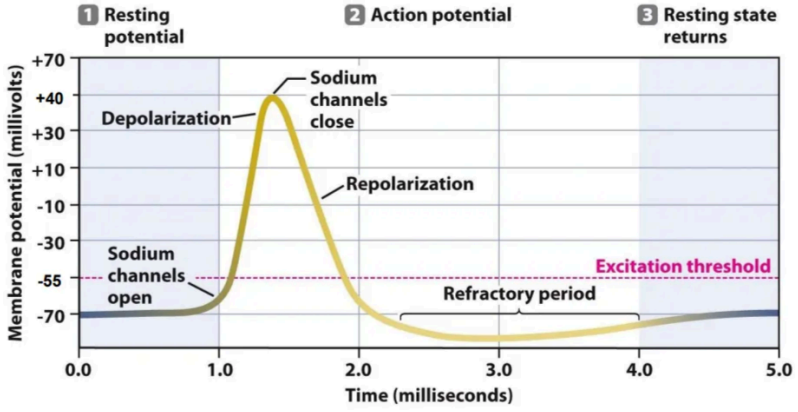
Neurons



Action potential: is a neural impulse, a brief electrical charge that travels down an axon

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Communication among neurons



Sodium enters the cell and potassium exits the cell

The amount of times you get an action potential tells you how strong it is

Neurons either fire or do not fire

- All-or-none law

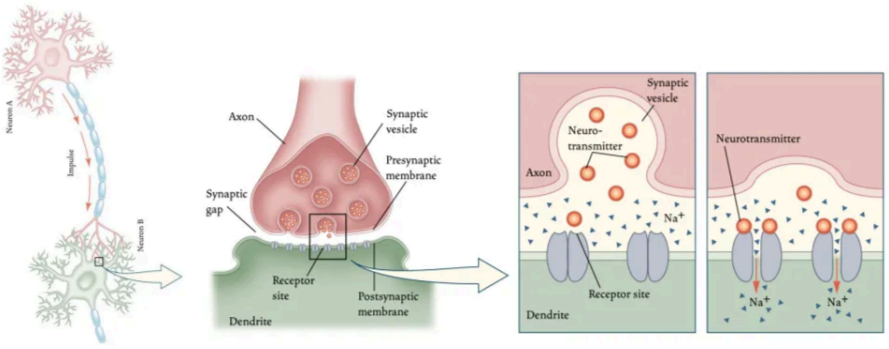
- Intensity variations by
  - the number of neurons firing (number)
  - the firing rate (frequency)

Neurons interact

- via synapses
- through chemical substances

Synapse

- the place where a signal passes from one nerve cell to another
- within synaptic vesicles you can find the neurotransmitters
- a receptor site needs a sufficient amount of neurotransmitters to make reach an action potential



Neurotransmitters

- chemical substances that transmit signals from one neuron to another
  - Lock-and-key Model
    - neurotransmitters can only bind with a particular receptor, like a lock and key
  - Effect is terminated by
    - autoreceptors (connecting to a pre-synaptic receptor instead of a post-synaptic one)
    - synaptic re-uptake (going back in the synaps it came out of)
    - enzyme deactivation (breaking)
  - The binding of a neurotransmitter with a receptor produces an excitatory or inhibitory signal

1. Neurotransmitters are made in the axon
2. Neurotransmitters are stored in vesicles
3. Action potentials cause vesicles to fuse to the presynaptic membrane and release their contents into the synapse
4. Released neurotransmitters bind to the postsynaptic receptors

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Drugs

- Agonists
  - increase the precursor (more neurotransmitters)
  - counteracting the cleanup enzymes (so they won't get deactivated)
  - blocking the re-uptake
  - mimicking the neurotransmitters action
- Antagonists
  - decrease precursor
  - increase effectiveness cleanup enzymes

- enhance the re-uptake
- blocking of receptors

Communication of the brain with the body

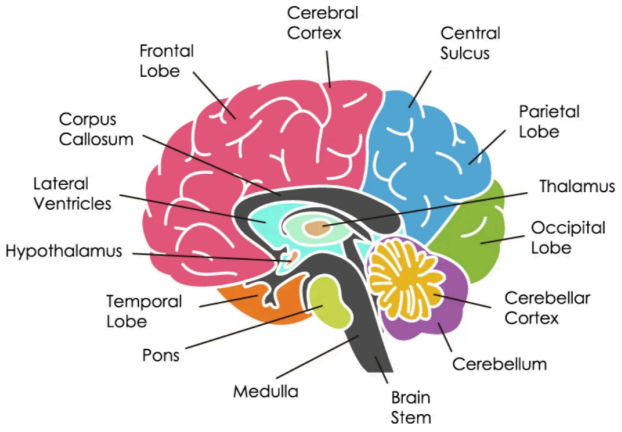
Nervous system

- Central nervous system
  1. Brain
  2. Spinal cord
- Peripheral nervous system
  1. Somatic nervous system
     - Skin muscles and joints send signals to the spinal cord and brain
     - Brain and spinal cord send signals to the muscles, joints and skin
  2. Autonomic nervous system
     - Glands and internal organs send signals to the spinal cord and brain
     - Brain and spinal cord send signals to the glands and internal organs
       - Sympathetic nervous system (prepares the body for action for stressful situations)
       - Para-sympathetic nervous system (returns the body to a resting state for normal, non stressful situations)

Endocrine system

- releasing of hormones

Brain



Cerebral Cortex - thought, planning

Cerebellum - motor function

Cerebellar cortex - coordinating movement

Central sulcus - dividing brain's motor and sensory areas

Hypothalamus - regulates body function

Corpus Callosum - connecting the two hemispheres

Lateral ventricles - protecting the brain

Thalamus - sensory gateway

Frontal lobe - personality, planning, emotion, problem solving

- Output: motor cortex - movement
  - left section controls the body's right side and vice versa

- Input: sensory cortex
  - left section receives input from the body's right side and vice versa
- Broca's area - speech production

Parietal lobe - touch, attention (movie: brain on fire)

Temporal lobe - hearing

- Wernicke's area - language comprehension

Occipital lobe - vision

Brain stem - survival

- Midbrain
- Pons
- Medulla

9th of September

Topic 3 - Consciousness

History of Consciousness

Psychology began as a science of consciousness

Behaviorists alienated consciousness from psychology, but after 1960, mental concepts (consciousness) started re-entering psychology

- Thanks to information processing technology

Modern definition

Subjective awareness of ourselves and our environment

An awareness of the sensations, thoughts, and feelings that one is attending to at a given moment, you're more controlled/focused when learning a complex concept of behavior. It becomes automatic by practice

Consciousness: one's subjective experience of the world, resulting from brain activity

- The brain and the mind are inseparable
- Each of us experiences consciousness personally
  - We cannot know if two people experience the world in exactly the same way: *explanation gap*

Cognitive Neuroscience

There is also mental activity if you think about doing something, or if you close your eyes and imagine something, the same parts of the brain are activated as if you actually see or do something

- This is why there can be responsiveness in comatose patients

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Divided consciousness

Information from the left visual field comes in it goes to the right side of the brain and vice versa

Consciousness arises as a function of which brain circuits are active

→ Global Workspace Model

- For patients with a split brain, the left side of the brain only comprehends what's seen on the right vision field and vice versa, except the right side of the brain can not comprehend as much as the left side, just

the basics.
- They can do and see two things at the same time, 'cause their brain works separate from each other, but if you ask them about what they saw they will only tell you about their right vision field, because only the left hemisphere has language abilities, the right side has limited language
  - This means that those people can perform actions they cannot verbally explain

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Variations in consciousness

Attention

- Refers to the process that enables you to focus selectively on some things and avoid focusing on others
  - Rapid toggling between activities makes you lose your attention
- Sometimes the unattended information breaks through, the ability to attend selectively to only one voice among many
- Unattended information may sometimes affect behavior → subliminal perception

Sleep

- You're alertness has gone all the way to zero, you're body temperature drops but you're secretion of growth hormones increases during sleep.
- Dreams: products of an altered state of consciousness in which images and fantasies are confused with reality
  - Non-REM dreams: dull
  - REM dreams: bizarre and intense

Drugs

- Tolerance: increasing amounts of drug needed to achieve the intended effect
- Addiction: drug use that remains compulsive despite its negative consequences (physical and psychological dependence)
- Withdrawal: physiological and psychological state characterized by feelings of anxiety, tension, and cravings for the addictive substance
- Stimulants: drugs that increase behavioral and mental activity and activate the sympathetic nervous system
  - amphetamines, methamphetamine, cocaine, nicotine, caffeine
- Depressants: reduce behavioral and mental activity by depressing the central nervous system
  - alcohol, anti-anxiety drugs - benzodiazepines, barbiturates
- Opiates (narcotics): depress or slow down the central nervous system; relieve pain and suffering
  - heroin, morphine, codeine
- Hallucinogens (psychedelics): produce alterations in cognition, mood, and perception
  - LSD, mescaline (from Peyote cactus), psilocybin mushrooms

11th of September

Topic 4: Sensation and perception - vision

The senses

- The underside of our brain's right hemisphere helps us recognize a familiar human face as soon as we detect it
- Human ears are most sensitive to sound frequencies that include human voices, especially a baby's cry

- Frogs, which feed on flying insects, have cells in their eyes that fire only in response to small, dark, moving objects

Sensation - the detection of external stimuli and the transmission of this information to the brain

Perception - the processing, organization, and interpretation of sensory signals (always an interpretation)

Distal stimulus (thing in the outside world) → Proximal stimulus (image on your retina) → Transduction (what the receptors do, translation of a physical stimulus from the outside world) → Sensation → Perception

Different senses:

- Vision
- Hearing
- Taste
- Smell
- Skin senses (pressure, temperature, pain)
- Vestibular sense
- Kinesthesia (feedback of body movement)
- Proprioception (feedback of body position?)
- Interoception (input from internal organs)

Thresholds

We have very restricted awareness:

- Absolute threshold
  - The minimum stimulus energy needed to detect a particular stimulus 50 percent of the time
    - Can I detect this sound?
  - Subliminal
    - Below absolute threshold
    - If you cannot detect it consciously at least 50% of the time, it is subliminal for you
- Difference threshold
  - Detecting small differences between stimuli
  - The minimum difference between two stimuli required for detection 50% of the time
  - We experience the difference threshold as a just noticeable difference
  - Weber's law:
    - The more intense the stimulus, the bigger the difference needs to be

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Quantitative variations (intensity)

- Rates of neuron firing
- Total number of neurons triggered

Qualitative variations (sensory quality)

- Different sensory qualities are signaled by different neurons
- Certain sensory qualities arise because of different patterns of activation across a whole set of neurons

Sensory Adaptation

- The tendency to respond less to a stimulus that has been present and unchanging for some time
- It is at the level of sensory receptors (no response)
- Benefit:
  - Freedom to focus on informative changes in our environment

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Vision - the retina

Light (direct or reflected)

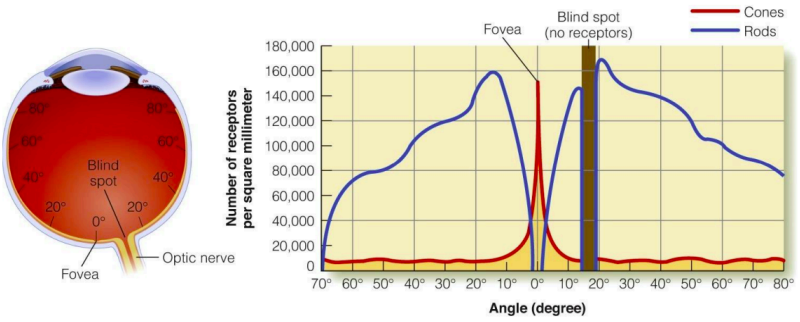
- Can vary in intensity and wavelength
- Is absorbed by pigments in the receptors (rods and cones) which transduce light energy into electric energy
    - this electric energy is carried to the brain via ganglia cells

Stimulus (light) → cornea → pupil, iris → lens - retina → optic nerve → brain's visual cortex

- Rods - night vision, no colour vision
- Cones - day vision, colour vision

Distribution of rods and cone across the retina

- higher resolution in the fovea



Colour vision

1. Trichromatic Theory (mix of RGB)
2. Opponent-process theory
    - there is no such thing as blue-yellow or red-green, they're opposites just like black-white
- These two theories represent the different stages of visual processing

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Vision - beyond the retina

- Damage in the temporal lobe ensures that you cannot discriminate objects, you don't see a difference
- Damage in the parietal lobe ensures that you cannot see the difference in space between objects (spacial awareness)

Two systems:

- "where" (path to parietal cortex, dorsal)
- "what" (path to temporal cortex, ventral)
- There is without a doubt a strong modularity (able to work individually but also together) between within each system

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Visual perception

The visual system is not a passive receiver of visual information but modulates, organizes, and interprets this information

Perception occurs on the basis of

- Bottom-up (data driven) processes
    - starts at your sensory receptors and works up to higher levels of processing
- Top-down (knowledge driven) processes

- Ex. priming, context etc.
- Constructs perceptions from this sensory input by drawing on your experience and expectations

Context

Our immediate context, and the motivation and emotion we bring to a situation, also affect our interpretations

- People's expectations influence their perceptions constantly
- Emotions can shove our perceptions in one direction or another
- Perception is influenced by culture

Experience → concepts, schemas

- Helps organize and interpret unfamiliar information
  - Apply top-down processing to interpret ambiguous sensations

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Perceptual organization

- Segregation of figure and ground
  - Lower areas are more often perceived as figure than higher areas
  - Convex forms are more often perceived as figure whereas concave forms are more often perceived as background
  - Perceptual interpretation on the basis of organizing principles
    - Ex. proximity (things that are closer are often grouped as if they belong together)
    - Ex. similarity (things that have the same shape and together are grouped as one, whereas different shapes that are together aren't)
    - Ex. continuity (things that are long and crossing over other things are still grouped as one thing, like cables)
    - Ex. closure (things that are supposed to be one shape with gaps are still considered as one shape)

Perception of Depth

Depth cues

- Binocular cues (uses both eyes to perceive the world in three dimensions)
  - binocular disparity (slightly different images captured by each eye due to their physical separation)
  - convergence (the inward turning of the eyes as they focus on a nearby object)
- Monocular cues (uses a single eye to perceive depth and distance
  - occlusion
  - linear perspective
  - texture gradients
  - shadow
  - motion parallax

Perception of size

Size perception depends on perceived distance

Perceptual constancies

- We correctly perceive different properties of distal stimuli despite continuously changing proximal stimuli
  - size constancy (we know someone is not really smaller is it's further away)
  - shape constancy
  - lightness constancy
  - color constancy

16th of September

Topic 5: Learning

Nonassociative Learning - learning about a stimulus, such as sight or sound, in the external world

- Habituation - when our behavioral response to a stimulus decreases
  - Allows a better ability to focus on what is new by ignoring that which has already become familiar
- Dishabituation - an increase in a response because of a *momentary* change in something familiar
- Sensitization - when our behavioral response to a stimulus increases
  - It's a more generalized reactivity
- Desensitization - is the extinction of sensitization

Associative learning - learning the relationship between two pieces of information

- Classical conditioning - a form of learning in which a neutral stimulus comes to elicit a response when it is associated with a stimulus that already produces that response
  - First order conditioning
    - When a CS (Conditioned Stimulus) produces a CR (Conditioned Response)
    - Pavlov, already reacting at ringing the bell, without seeing the food, because they know they'll get food after the ringing
  - Second order conditioning
    - When a CS is paired with a new S, the new S produces a CR
    - When you turn on the lights before ringing the bell before giving the food every time, the response will already start at the lights being turned on
  - Stimulus generalization
    - The CR can also be elicited by stimuli that are similar to the CS
  - Conditioning requires
    - contiguity (together in time) - such that learning only occurs when the CS is presented before the US (Unconditioned Stimulus)
    - contingency - the CS should be informative with respect to the chance that the US is offered, it doesn't have to be all or nothing, when something happens 40% of the time it's already enough to elicit a CR
  - Cognitivist view:
    - The CS functions as a sort of predictor for the US
    - The CR is a reaction in preparation of the US
- Operant conditioning - when we learn that a behavior leads to a certain outcome a.k.a. instrumental conditioning
  - the consequence of an action/response determines the likelihood that it will be performed again

        - the relationship between response and the consequence is learned (reinforcer/reward or punishment)
        - performance is strengthened when followed by a reward and weakened if it's not
    - Reinforcers
        - Primary reinforces - food , water etc.
        - Secondary reinforces - money etc.
        - Positive reinforcement - adding something (getting a reward)
        - Negative reinforcement (still good, because reinforcement is good) - taking something away (taking pain away)
        - Continuous reinforcement - every time they get the reward
        - Partial reinforcement (Schedules of reinforcement) - they get the reward sometimes, not every time
            - Fixed ratio - after a standard amount of responses (after 4 times)
            - Variable ratio - after an average of responses (after an average of 4 times, could also be 1 or 8)
            - Fixed interval - after a response after a set time (1st response after 4 minutes)
            - Variable interval - after a response after an average time (1st response after on average 4 minutes, could also be 1 or 8 minutes)
            - Partial reinforcement extinction effect
                - partial reinforcement leads to less extinction than continuous reinforcement, because it doesn't get rewarded every time so they think there's always a chance
    - Punishment
        - Positive (aversive) punishment - adding a stimulus (getting smacked)
        - Negative punishment (response cost) - removing a stimulus (no TV for a week)
    - Conditioning requires
        - Contiguity (together in time) - learning only occurs when the reinforcer or punishment immediately follows the response
        - Contingency - the reinforcer or punishment should occur more likely after the response than otherwise.
            - Without contingency (so without a consequence to their actions, whether that is a reward or a punishment) they get helpless (learned helplessness)

Observational Learning (social learning) - learning by watching how others behave

- Modeling
    - The acquisition or modification of a behaviour after exposure to another individual performing that behaviour
    - Basically just imitation
- Vicarious learning
    - When you imitate behaviour based on someone else's reinforced behaviour
    - Imitating someone because they got rewarded

18th of September

Topic 6: Memory

The study of memory

Perception

- the organization of current stimuli

Memory

- the organization of stimuli from the past (ideas and events)
- the nervous system's capacity to retain and retrieve skills and knowledge
- we have multiple memory systems, and each memory system has its own "rules"

Memory operates over time in three phases in a sequential process

- Encoding
- Storage
- Retrieval

Sensory memory

Feeds our active working memory, recording momentary images, sounds and strong scents

But sensory memory is like a lightning flash, it's fleeting

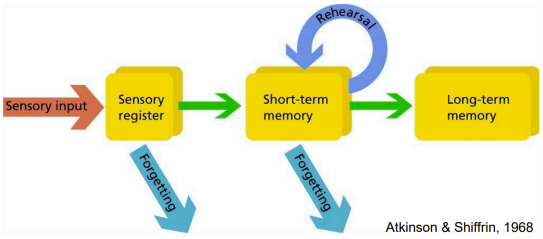
- Iconic memory
  - A momentary sensory memory of visual stimuli; a photographic or picture-image memory lasting no more than a few tenths of a second
- Echoic memory
  - A momentary sensory memory of auditory stimuli; if attentions is elsewhere, sounds and words can still be recalled within 3 or 4 seconds

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1. Encoding

Good encoding provides an effective means for retrieval later

- Stage theory of memory



- STM (Short Term Memory)
  - Limited capacity
    - Memory span task: magic number seven
      - Items disappear as a result of decay and replacement
    - By recording (chunks) → expansion of capacity
- LTM (Long Term Memory)
  - Unlimited capacity
- Through rehearsal STM → LTM

Evidence for two memories (STM and LTM)

- Primary Effect
  - memories from the long term memory
  - more accurate than the middle, but less accurate than recency
- Recency Effect
  - memories from the short term memory
  - way more accurate in your memory because you've just now seen them

Modern memory theory

- STM → working memory
  - Information one is thinking about
- LTM
  - Information, stored during longer intervals, which is currently not actively used
- Not everything has to go through your STM to get in your LTM, it can go from your sensory memory immediately to your LTM
- Rehearsal also doesn't make STM into LTM, most of the time it stays in your STM

Depth of processing

- Shallow processing (surface form)
- Deep processing (meaning)

Understanding

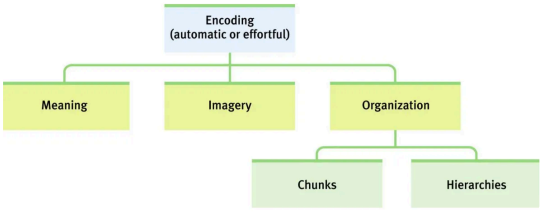
- A schema may help you to understand and subsequently remember information contents

Effortful processing

- Requires attention and conscious effort (ex. your new pinpass code)

Automatic processing

- Unconscious encoding of incidental information (ex. space, time and frequency)
- And of familiar or well-learned information (ex. sounds, smells, word meanings and the way to school)



Encoding meaning (semantic encoding) results in better recognition later than visual or acoustic encoding

- ex. Don't just repeat the word patella over and over again but search the meaning: kneecap

Mental pictures (imagery) are a powerful aid to effortful processing, especially when combined with semantic encoding

- ex. Showing anorexia in a picture may be more powerful than simply talking about it

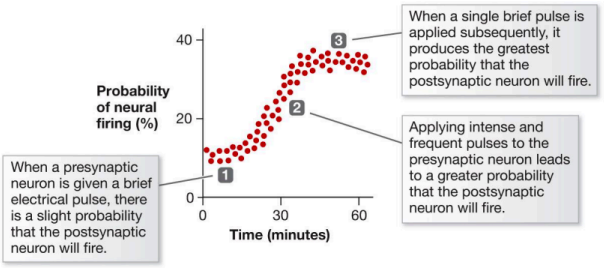
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2. Storage and consolidation (in memory traces)

After encoding → storage in memory traces

- Memory consolidation
  - the neural process by which encoded information becomes stored in memory → transient and fragile potential memory contents are transformed into a more permanent state
  - Long-term potentiation (LTP)

        - Strengthening of a synaptic connection, making the postsynaptic neurons more easily activated by presynaptic neurons
    - After rehearsal -> reconsolidation (might corrupt)
        - It's vulnerable for change every time a memory is retrieved, even if it's already been consolidated



Hippocampus

- a very important part for LTM
- if damaged it disrupts the formation and recall of explicit memories
- Memory consolidation
    - the neural storage of a long-term memory
    - sleep supports this

Stress hormones & memory

- Heightened emotions (stress-related or otherwise) make for stronger memories.
- Flashbulb memories are clear memories of what we were doing during emotionally significant moments or events. (9/11)
- Stress hormones make more glucose energy available to fuel brain activity

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Varieties of memory

Explicit memory

- Requires conscious effort and often can be verbally described
- Retention of facts and experiences that we can consciously know and "declare"
- Declarative memory

1. Episodic memory: Personally experienced events
2. Semantic memory: Facts and knowledge

Implicit memory

- Does not require conscious effort and often cannot be verbally described
- Retention of learned skills or classically conditioned associations independent of conscious recollection
    - ex. you know Paris is the capital of France but don't remember when you learned that fact
- Nondeclarative memory

1. Classical conditioning: Associating two stimuli elicits a response
2. Procedural memory: Motor skills and habits

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1. Retrieval

- Recognition
    - The person must identify an item amongst other choices (ex. MC tests)
- Recall
    - The person must retrieve information using effort (ex. fill in the blank tests)

Good retrieval cues

- re-create the context in which the original learning occurred → context reinstatement (ex. divers)
- re-create the state in mind in which the original learning occured
- a richer encoding context results in better recall
  - encoding specificity principle

Forgetting

An inability to retrieve information due to poor encoding, storage, or retrieval

- Encoding failure
  - We cannot remember what we do not encode, information never enters long-term memory (ex. absentmindedness, memory decay)
- Absentmindedness: the inattentive or shallow encoding of events
  - During most of our daily activities we are consciously aware of only a small portion of both our thoughts and our behaviours
- Memory decay: forgetting over time (forgetting curve)
  - Decay of memory traces with time (causes by metabolic processes)
  - Interference
    - After sleep less forgetting than after wakefulness
    - Proactive interference (study psychology → study anthropology → take anthropology test)
    - Retroactive interference (study psychology → study anthropology → take psychology test)
- Memory reconstructions (distortions of memory)
- Amnesia: a deficit in long-term memory resulting from disease, brain injury, or psychological trauma in which the individual loses the ability to retrieve vast quantities of information
- Memory bias
  - the changing of memories over time so that they become consistent with current beliefs or attitudes
  - flashbulb memories can be biased
- Source misattribution
  - memory distortion that occurs when people misremember the time, place, person, or circumstances involved with a memory
- Suggestibility
  - the development of biased memories from misleading information
    - different wordings of questions after an event may alter the participant's memories of the event
  - memories can be distorted, or even implanted, with false information
  - memories can be altered by the intrusion of generic knowledge or semantic associations
- Misinformation effect
  - while tapping our memories, we filter or fill in missing pieces of information to make our recall more coherent
  - incorporating misleading information into one's memory of an event
  - after exposure to subtle misinformation, many people misremember
  - guessed details are absorbed into our memories

23rd of September

Topic 7: Thinking and language

Mental representations

- Analogical representation
  - representing something in your mind the way it is in the real world (you see a cat)
  - they're dissimilar from pictures
- Symbolic representation
  - representing something with symbols (you see the word 'cat')

  Categorization
  - Prototype theory
    - an 'average' representation of the most common members (can vary in strength, depending on representational similarity) of a category
  - Exemplar theory
    - category membership depends on similarity to exemplars
      - typical: similar to many earlier examples that you have seen
      - atypical: similar to few examples you have seen

Decision making

selecting one option among several possibilities

- Judgement
  - Deciding on the likelihood of various events using incomplete information
- Rational search
  - A single "correct" choice after a cognitive evaluation

Normative decision theories define how people should make decisions

- Expected utility theory
  - Decisions should be made to maximize the outcome
  - This should depend on
    - the values of the alternative options
    - the probabilities associated with the alternative options
    - ex. buying a lottery ticket for 10 euro when the reward could a million

Descriptive decision theories indicate how people actually make choices

- Heuristics
  - shortcuts (rules of thumb or informal guidelines) used to reduce the amount of thinking that is needed to make decisions
    - heuristic thinking often occurs unconsciously (based on feeling)
    - heuristic thinking allows us to decide quickly
    - heuristics can also result in errors or faulty decisions
  - Availability heuristic
    - deciding based on the answer that most easily comes to mind
      - ex. death by a car accident vs. brain hemorrhage
  - Representativeness heuristic
    - deciding based on how typical an individual, object or event belongs in a certain category

    - Affective forecasting heuristic
        - the tendency for people to overestimate how events will make them feel in the future
    - The paradox of choice
        - when too many options are available, especially when all of them are attractive, people experience conflict and indecision
            - picking what to eat from a menu
- Anchoring
    - the tendency, in making judgements, to rely on the first piece of information encountered
    - ex. 50% sale, so you think it's cheap without knowing what the product is really worth
- Framing
    - in decision making, the tendency to emphasize the potential losses or potential gains from at least one alternative
    - ex. winning 1000 euros feels way less important than losing 1000 euros

A good strategy is to satisfice rather than optimize your decisions

Satisficers vs. maximizers

Problem solving

Thinking process in which we move from an initial state to a goal state → goal-directed

- Obstacles to problem solving
    - Complexity
    - Mental set
        - instruction/practice ex. math problems
        - perceptual organization of the problem ex. drawing a house in one strike
        - functional fixedness ex
            - having fixed ideas about the typical functions of objects
- Overcoming obstacles to solutions
    - Understanding → formulate subgoals
    - Restructuring ex. using all you resources, sometimes we don't realize what we can use
    - Use an algorithm
    - Use an analogy
    - Wait for a sudden insight

Language

A system of communication using sounds and symbols according to grammatical rules

- Syntax: rules
- Semantics: meanings

As the brain develops, so does the ability to speak and form sentences

- development occurs in an orderly way
- listening preferences in newborns are influenced by the language or languages spoken by mothers during pregnancy
- newborns can discriminate all the phonemes (klanken) but lose this ability after 6 months

The human vocal tract has the ability to make many more sounds than any language uses

- language differ from one another not only by the words that are used but also by the number of phonemes and the patterns of morphemes (stamwoorden?)

Aphasia

- a language disorder that results in deficits in language comprehension and production
- Expressive aphasia (Broca's aphasia)
  - interruption of ability to speak
- Receptive aphasia (Wernicke's aphasia)
  - difficulties to understand the meaning of words

25th of September

Topic 8: Intelligence

Francis Galton (1822–1911)

- Statistical analysis of mental tests
  - measure individual achievement on mental exercises

Pearson (1857–1936)

- Covariation of multiple traits
  - mathematical possibility of predicting capacity on the basis of many tests of various mental traits

Spearman (1863–1945)

- Intelligence: a single generical factor and some specific traits

L. L. Thurstone (1887–1955)

- Factor analysis
  - Made possible the development of a combined prediction method for individual abilities

Alfred Binet (1857–1911)

- Assessing the intellectual aptitude of schoolchildren
- The concept of mental age

William Stern (1871–1938)

- Dividing the mental age by chronological age to calculate an individual's intelligence quotient (IQ)

*In 1916, re-standardized the Binet test for American use

*In 1917, intelligence testing for the army adopted it as a selection device

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What is intelligence?

"The ability to reason, plan, solve problems, think abstractly. comprehend complex ideas, learn quickly, and learn from experience"

~ Gottfredson, 1997

"the ability to use knowledge to reason, make decisions, make sense of events, solve problems, understand complex ideas, learn quickly, and adapt to environmental challenges"

~ Gazzaniga, 2018

Intelligence Testing

- Standardized psychometric tests
  - IQ = (MA/CA) * 100 (Binet, 1911)

        - MA = Mental Age
        - CA = Chronological Age
    - Deviation IQ
    - The Binet tests were originally for children
- Tests for adults
    - Wechsler Adult Intelligence Scale (WAIS) for example
        - verbal and performance tests, individual test
    - Scholastic Aptitude Test (SAT)
        - group test
    - Raven's Progressive Matrices Test
        - Non-verbal test
- General
    - High reliability
    - High predictive validity (depends on complexity of someone's job/task)
    - Some caveats (e.g. anchored in western society etc.)

Psychometric approach

- An attempt to understand the nature of intelligence by studying the pattern of results obtained on intelligence tests
    - Spearman: we have general intelligence but that consists of multiple specific abilities
    - Cattell: fluid vs. crystallized intelligence (your vocabulary won't just go away with time)
        - fluid intelligence drops with age
- Standardized intelligence tests fail at capturing other types of intelligence
    - Types of intelligence: Analytic, creative, practical, emotional intelligence etc.
- Different measures correlate with g
    - Simple RT (reaction time) and choice RT task
    - Discrimination task (is A different than B, higher g à shorter time)
    - Working memory capacity
    - Executive control (ability to switch tasks etc.)
- Score on intelligence tests is influenced by genetics and environment
    - Genetic influences
        - Twin studies
            - Identical: .86 correlation in intelligence à doesn't drop when raised apart
            - Fraternal: .6
    - Environmental influence
        - In rich environments => more variations compared to impoverished
            - Can try and fail, more opportunities etc.
- Heritability H
    - Between 0 and 1
    - H = Generic Variance / Phenotypic variance
    - 0 = everything is due to environment
    - 1 = everything due to genetics (like having 2 eyes)
    - Based on western middle class, so little reliability
    - Statistical estimate of the proportion of variance in a population
- Flynn effect:
    - Environmental changes change root score of IQ à average IQ now 107 instead of 100

        - More stimulation, better nutrition
- Now: reversed Flynn effect
- Group differences
    - Men vs women
        - Very hard to proof
        - Publication bias (want significant results)

Beyond IQ test
Relation IQ and cognitive performance
Roots of intelligence

30th of September

Topic 9: Scientific reasoning

The law of attraction: *"like attracts like"*

Hindsight Bias

- The "I-knew-it-all-along" phenomenon
- After learning the outcome of an event, many people believe they could have predicted that very outcome
- Anything seems common sense once explained

Overconfidence

- Sometimes we think we know more than we actually know

*Both hindsight bias and overconfidence leas us to overestimate our intuition

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Fixation of belief

- Belief
    - an attitude of some person toward a particular claim
    - ex. believing in the law of attraction, aliens etc.
- Claim (or statement)
    - true
    - false
- There is no direct connection between belief and truth
    - believing that a given claim is true does not provide information about the truth or falsity of the claim itself
- Why are some claims like "time dilution" scientifically justified and others like "aliens" aren't
    - the first claim is based on the proper use of the scientific method and the second one isn't
- What are methods to fix beliefs
    - Non-scientific methods
        - Personal experience
        - Intuition
        - Authority
    - Scientific methods
        - based on the basis of empirical research

        - it reaches conclusions by combining proper reasoning (logic arguments) and systematic empiricism (verifiable evidence)

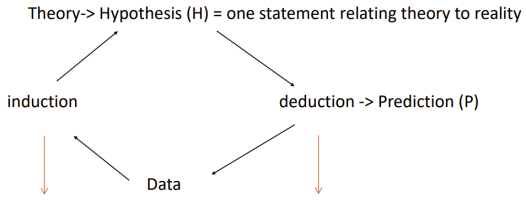
Solvable Problems

- Science only deals with solvable problems
- Researchers can investigate only those questions that are answerable, given current knowledge and research techniques
- To be considered scientific:
    - the questions addressed must be potentially solvable
    - observations must be systematic and empirical
    - research must be conducted in a manner that is publicly verifiable

*Pseudoscience involves evidence that masquerades as science but that fails to meet one or more of the three criteria used to define science

*Psychologists, like all scientists, use the scientific method to construct theories that organize, summarize and simplify observations.

Empirical cycle (theory-data cycle)



Induction

- specifying a general rule based on specific observations
    - ex. seeing a handful of white swans so concluding that every swan in that park is white even though you haven't seen the black swan
- the conclusion is often plausible but not inescapable
- Inductive reasoning
    - bottom-up, using observations (empiricism)
    - ex. infinity cannot (never) be proven because nobody can count to infinity

Deduction

- Specifying a conclusion about something specific on the basis of a general rule
- The conclusion follows inevitably from the assumptions/premises
- Deductive reasoning
    - top-down, using things you already know (rationalism)
    - ex. infinity exists, given the fact that you can always add a number to the previous number
    - Syllogism
        - 2 premises → conclusion
        - valid or invalid (not the same as true or false)
        - 2 types of syllogisms
            - Categorical (All A are B...)
            - Conditional (If A then B...)

Scientific Method

A method to provide good justification for the belief that a claim or statement (hypothesis) is true or false

- Goal of Science
  - To find good justification for the belief that a claim or statement is true or false
    - by means of a conditional argument
- Conditional argument
  - A = antecedent B = consequent
  - Affirming the antecedent: if it rains i stay at home + it rains = i stay at home
    - Valid argument
  - Denying the consequent: if it rains i stay at home + i do not stay at home = it's not raining
    - Valid argument
  - Denying the antecedent: if you are a proofreader you have a job + you're not a proofreader = you don't have a job
    - Not a valid argument
  - Affirming the consequent: If i live in Amsterdam I live in the NL + i live in the NL = i live in Amsterdam
    - Not a valid argument
- The type of argument used to justify or falsify a scientific hypothesis is the same across different scientific disciplines
  - The conclusion of the argument will be either the hypothesis or the negation
- Hypothesis
  - A testable prediction, often prompted by a theory, to enable us to accept, reject or revise the theory
- Research
  - Requires us to administer surveys or experiments
- Operational definition
  - a statement of procedures (operations) used to define research variable
  - ex. how you're going to measure aggression is your operational definition for aggression
- Replication
  - Repeating the essence of a research study to see whether the basic finding extends to other participants and circumstances
  - Usually with different participants in different situations

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Research Process

1. Theory: Social learning theory

lead to

2. Hypothesis

lead to

3. Research and observations

generate and refine, back to step 1

1. If (H and IC and AA) then P
2. If (not-H and IC and AA) then probably not-P

H = Hypothesis

IC = Initial Conditions (description of state at the beginning of the experiment)

AA = Auxiliary Assumptions (external, testable hypotheses)

P = Prediction (follows deductively from H and IC and AA)

Falsifiability criterion

People believe statements which

- can be demonstrated to be false
- can be demonstrated to most likely true
- cannot be demonstrated to be false or true

To find good justification for the belief that a hypothesis is true, the hypothesis needs to be potentially falsifiable

- has to be precise too

Falsifiability criterion

- If we test our ideas, we can know when we are wrong. And if we know when we are wrong, we can correct our mistakes. If we don't, we can't.

Two Theories

- The Evolution Theory of Darwin is a scientific theory because it generates specific predictions rendering the theory potentially falsifiable
- The Intelligent Design Theory is not a scientific theory because it does not satisfy the falsifiability criterion
  - However, everybody is free to believe in this theory
  - Believing in intelligent design is a matter of faith and has nothing to do with science
  - This theory has not been repeatedly submitted to scientific testing

An established theory in science is based on far more than faith

- A theory in a scientific sense is a conceptual structure that is supported by a large and varied set of data.

2nd of October

Topic 10: Validity

Independent variables

- Manipulated variables

Dependent variables

- Measured variables depending on the independent variables

Confounding variables

- Variables that change simultaneously with the independent variables
- ex. time of testing, order

Control variables

- Variables that are being held constant on purpose
- ex. number of words, colors

Random error

- ex. noise during testing

Conceptual variable

- A variable stated at an abstract (theoretical) level
- ex. anxiety

Operational variable

- The specific way in which a construct is manipulated or measured in a study
  - Assigning numbers is a good start
- ex. test score → control or anxiety group

Goals of Psychology

- To describe behavior and mental processes
- To predict behavior and mental processes
  - Mainly by way of assessing the relationship between two or more variables
- To explain and understand behavior and mental processes
  - Experimentation, this is the backbone of psychological research
  - Experiments isolate causes and their effects
- To influence/control behavior and mental processes

Three claims (statements)

- Frequency claims
  - how often something occurs
  - focuses on one variable, not measuring a relationship
- Association claims
  - focuses on the relationship between at least two variables
    - correlation
    - covariation
  - in this claim there is a relationship, which could mean that there is a third variable causing both to variables to change
- Causal claims
  - focuses on causation, a change in one variable is responsible for changing the value of another variable
    - experimental research
  - in this claim there isn't necessarily a relationship

Validity

- the appropriateness of a claim
- A valid claim is: reasonable, accurate and justifiable

Construct validity

- An indication of how well a conceptual variable is measured or manipulated in a study
- Threats to construct validity
  - inadequate operational definition
    - ex. does a test measure anxiety or absence of self-confidence?
  - mono-operation bias
    - ex. mental workload

External validity

- An indication of how well the results of a study generalize to, or represent, individuals, settings, places, and times (contexts) besides those in the study itself (the ones you want to make a claim about)
- Threats to external validity
  - Selection biases
    - ex. study on the effect of marijuana on the ability to solve math problems
  - Study setting is different from other settings
    - ex. covid spread in urban vs rural areas
  - Specific time at which the study is performed is different from other times
    - ex. study on attitude concerning eating meat after BSE (mad cow disease) scandal
- Random assignment
  - Assigning participants to experimental and control conditions by chance
  - Minimizes pre-existing differences between those assigned to the different groups

Internal validity

- The extent to which it is possible to rule out alternative explanations for a causal relationship between two variables
- Threats to internal validity
  - Maturation
  - History
  - Testing
  - Instrumentation
  - Regression to the mean
    - expected return to the average, even if they were really bad in the pretest, other random reasons made sure they do better than last time
  - Attrition/experimental mortality
    - difference between pretest and posttest group because some dropped out
  - Selection effects
  - Design confounds
    - the presence of another variable that unintentionally varies systematically with the independent variable
  - Observer bias
    - single and double-blind studies
    - no one should know in who is in what group, not the people in the experiments and not the judges or observers
  - Demand characteristics
  - Placebo effects

Statistical validity

- The extent to which a study's statistical conclusions are accurate and reasonable
  - How well do the numbers support the claims?
- Threats to statistical validity
  - Violated assumptions of the test statistics
  - Fishing and the error rate problem
  - Low statistical power

7th of October

Topic 11: Measurement, reliability, and descriptive research

Measurement

A systematic way to assign numbers or names to objects and their features

- Nominal scale - differences (categorical), they are only named
- Ordinal scale - ranked order
- Interval scale - equal intervals (add/subtract)
- Ratio scale - absolute zero's (ratio's)

Importance of measurement scales

- determines the possible statistical operations
- determines the possible conclusions

Reliability

Consistency of measurements

Observed score = true score + error score

- the measured value

True score

- the real value

Error score

- the difference between the measured value and the real value which is affected by
  - the test (situation)
  - the person

Different types of reliability

- Test reliability
  - Test-retest reliability
  - Internal reliability
    - Split-half reliability
    - Cronbach's alpha
- Interrater reliability
- Experimental reliability

Descriptive research

???

9th of October

Topic 12: Correlations and Experiments

Correlational research

- Based on association claims
- Measure not manipulate variables
- Correlation coefficient (r)
  - quantify relationship
  - product-moment correlation coefficient or Pearson's r
  - measure indicated linear relationship between variables (negative or positive)
  - varies between -1 and +1
    - absolute value = strength
    - sign (+/-) = direction
  - statistical validity: does data support the claim
    - statistic validity and significance often correlate
    - outlies can affect correlation
    - is correlation statistically significant
      - not reject null hypothesis (null hypothesis = there is no difference): alpha > 0.05 (or 0.01/0.001)
      - reject null hypothesis (and accept your own): alpha < 0.05 (0.01/0.001)
  - directionality problem: we don't know what direction the correlation is (e.g. which one causes which)
  - 3rd variable problem à spurious correlation (e.g. correlation between ice cream eating and drowning)
  - Covariance: are variables correlated?
  - Temporal precedence: which came first (can't be solved with correlation)
    - Cross-legged-panel-procedure (way to use correlation for that)
      - Measure things with a lot of time apart and see if results are the same; shows time frame of what happened first
  - Internal validity (3rd variable problem) (can't be solved by correlation)
    - Way around it: Separating participants into subgroups on the basis of reasonable 3rd variable
      - If correlation in subgroups disappears with 3rd variable than 3rd variable is important
  - Sampling important
  - Correlation with r only suggest linear relationships, not, for example, curvilinear associations
  - Internal validity is low because no manipulation of variables

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Basis of experimenting

- Null results (H0)
  - Independent variable did not affect dependent variable
  - Manipulation was too week

    - Insensitive measures
    - Unreliable measurements
        - Too much variability in data à bigger error bars
    - Ceiling effect
    - Floor effect
    - Not enough power (small n)

Experimental research

- Remember that randomization of groups is important
- Assigning subjects into different levels of independent variable(s)
    - Between-subject design (independent groups design)
        - Separate groups receive different levels of independent variables
        - Has control group
    - Within-subjects design (within-groups/repeated-measures design)
        - All subjects receive all levels of the independent variable(s)
        - Has control condition
        - ø